



Session Initiation Protocol

ITAG Lunchtime Seminar

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Outline

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- SIP Overview
- IS SIP Technology Trial
- Telephony Futures
- Play Time



What's SIP

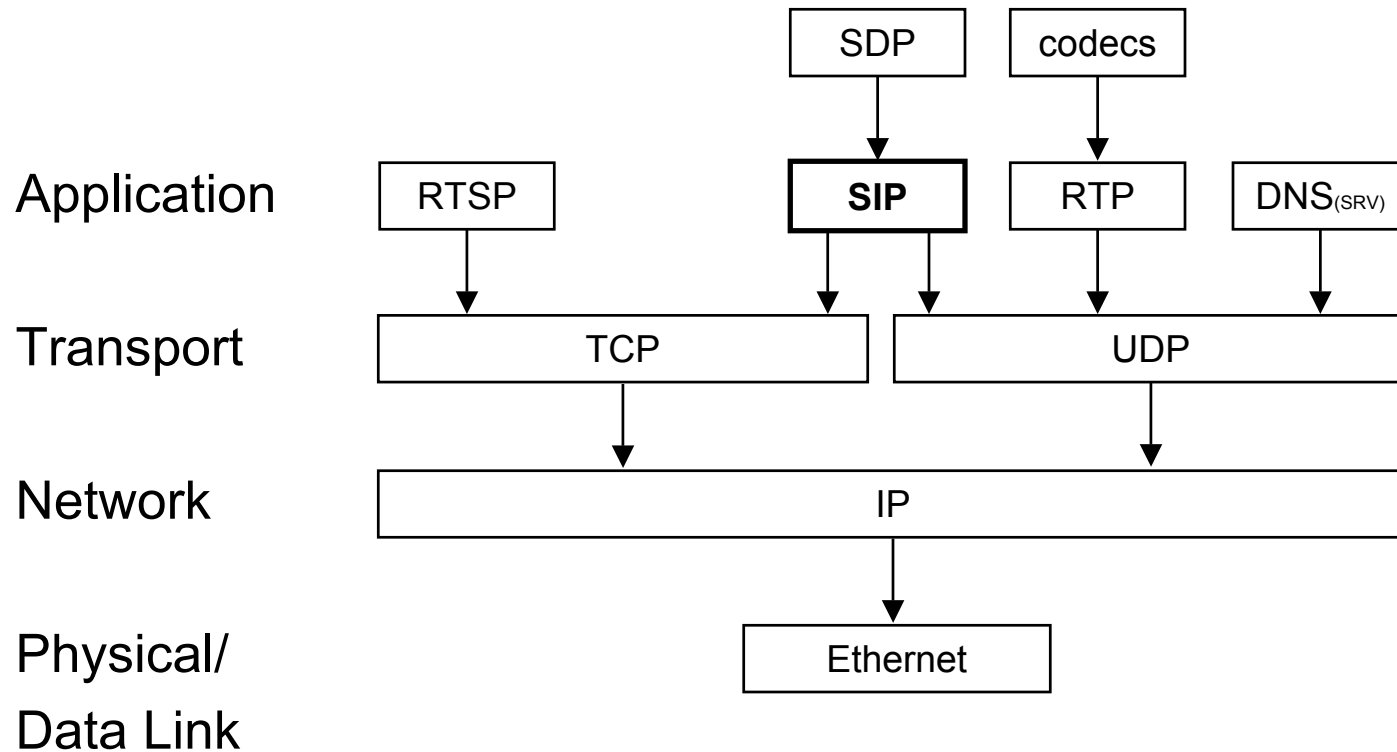
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- IETF RFC 3261
 - Replaces RFC 2543
- “The Session Initiation Protocol (SIP) is an application-layer control (signaling) protocol for creating, modifying and terminating sessions with one or more participants.”
- Can be used for voice, video, instant messaging, gaming, etc., etc., etc.
- Follows on HTTP
 - Text based messaging
 - URIs – ex: sip:dbaron@MIT.EDU



Where's SIP

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SIP Components

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- User Agents
 - Clients – Make and receive “calls”
 - Servers – Accept requests
- Server types
 - Redirect Server
 - Proxy Server
 - Registrar Server
 - Location Server
- Gateways



SIP Methods

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- INVITE Requests a session
- ACK Final response to the INVITE
- OPTIONS Ask for server capabilities
- CANCEL Cancels a pending request
- BYE Terminates a session
- REGISTER Sends user's address to server



SIP Responses

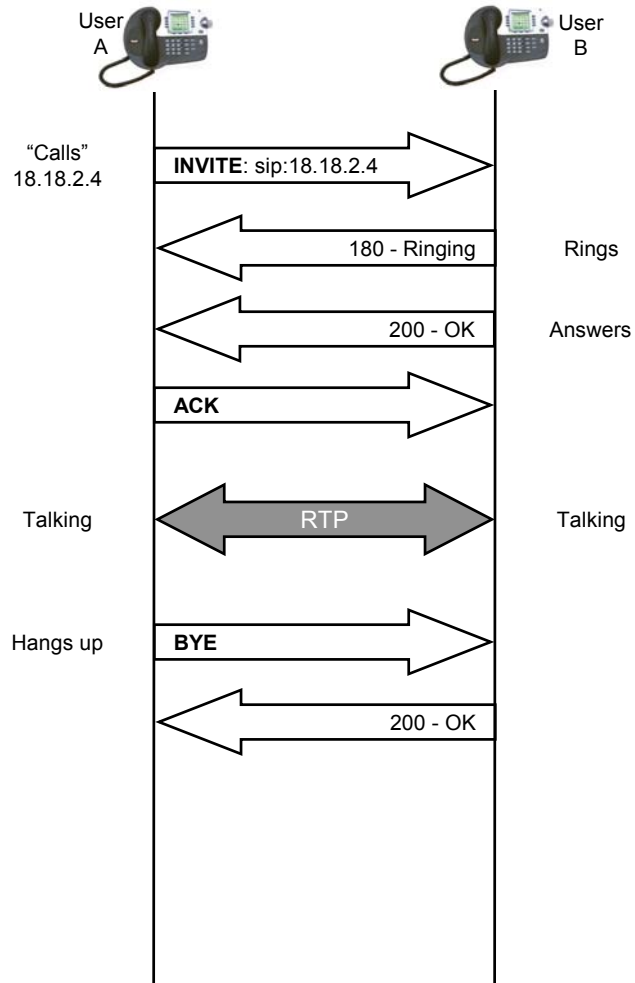
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|-------|----------------|-----------------------|
| • 1XX | Provisional | 100 Trying |
| • 2XX | Successful | 200 OK |
| • 3XX | Redirection | 302 Moved Temporarily |
| • 4XX | Client Error | 404 Not Found |
| • 5XX | Server Error | 504 Server Time-out |
| • 6XX | Global Failure | 603 Decline |



SIP Flows - Basic

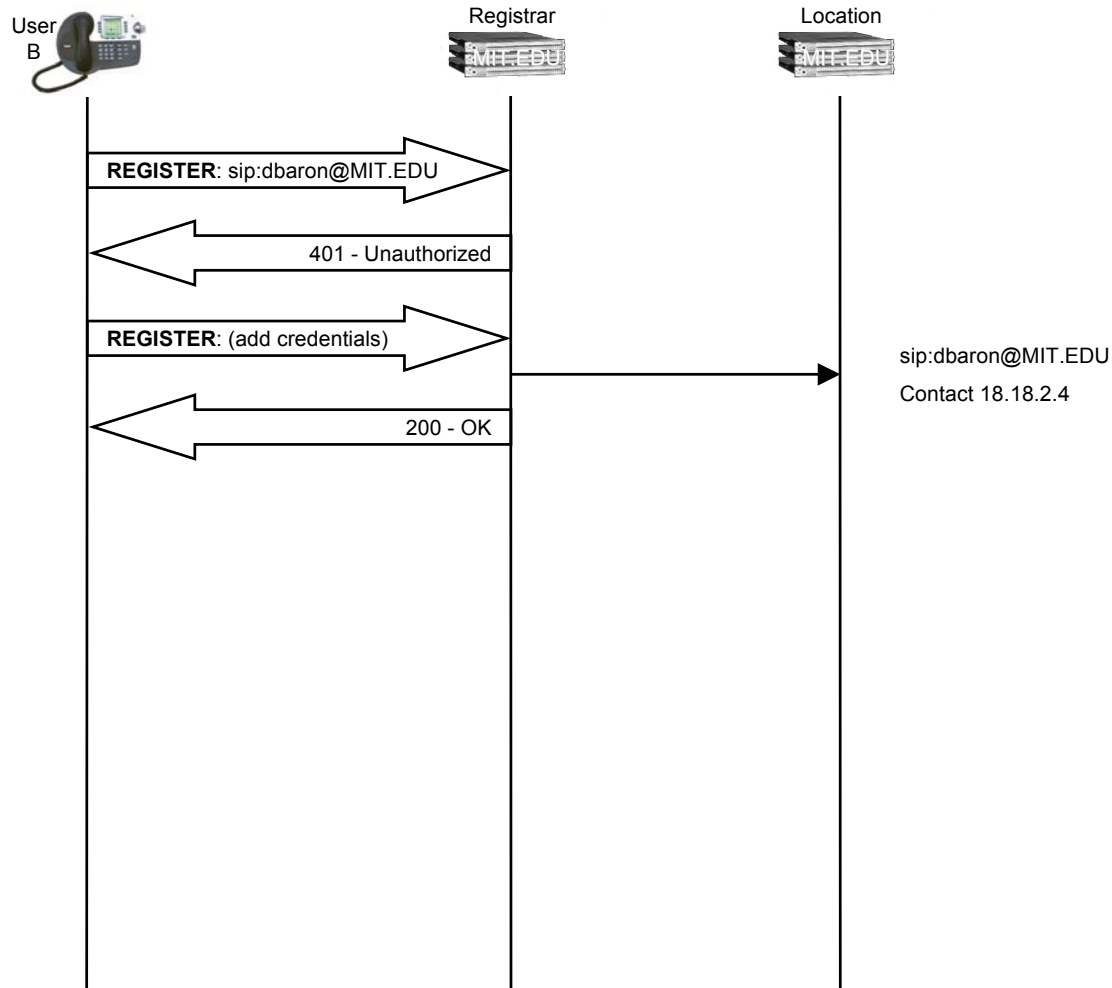
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SIP Flows - Registration

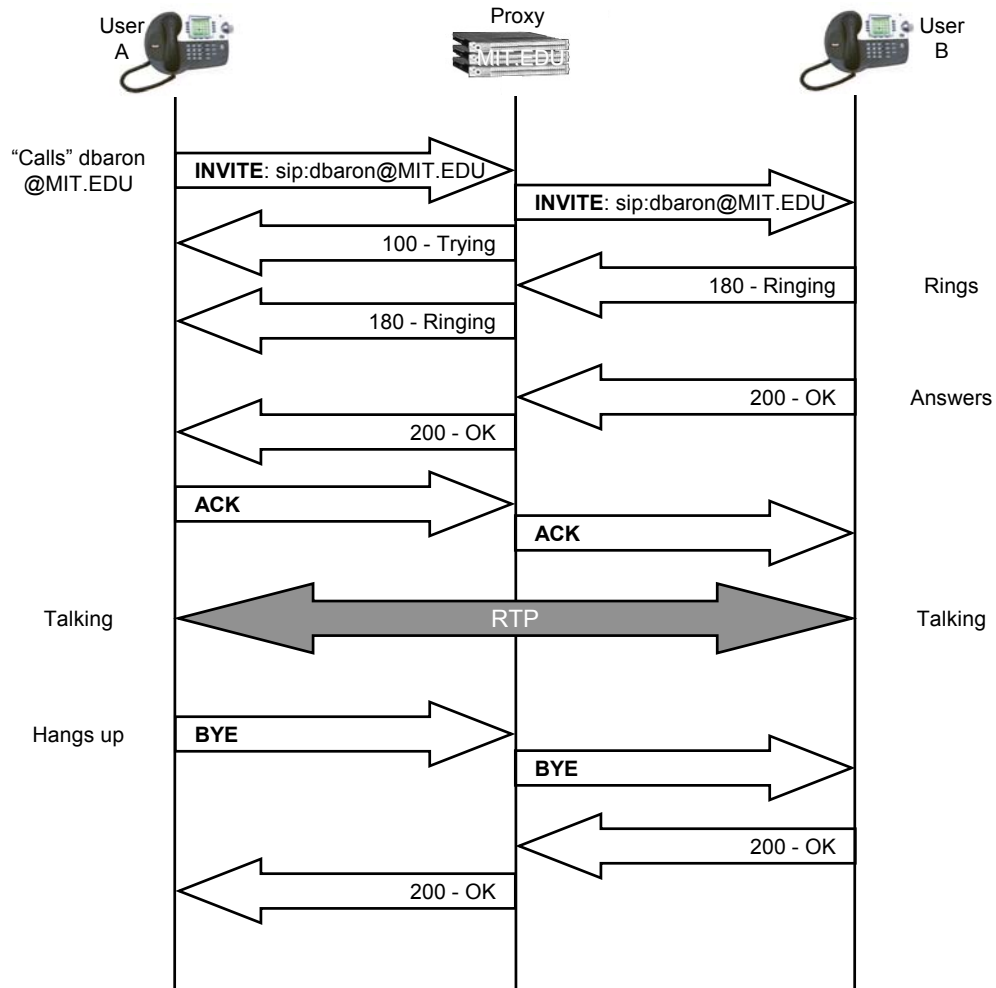
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SIP Flows – Via Proxy

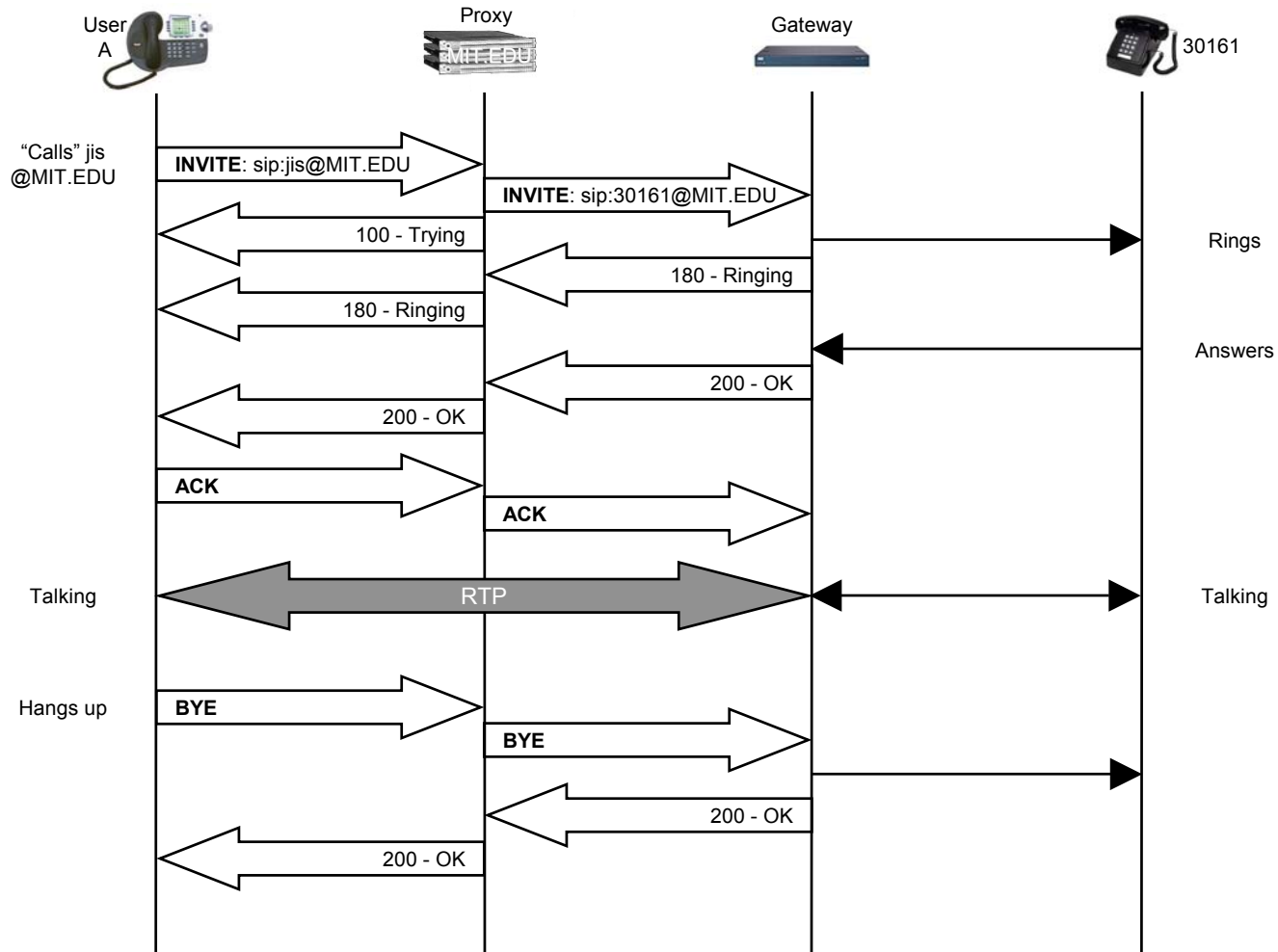
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SIP Flows – Via Gateway

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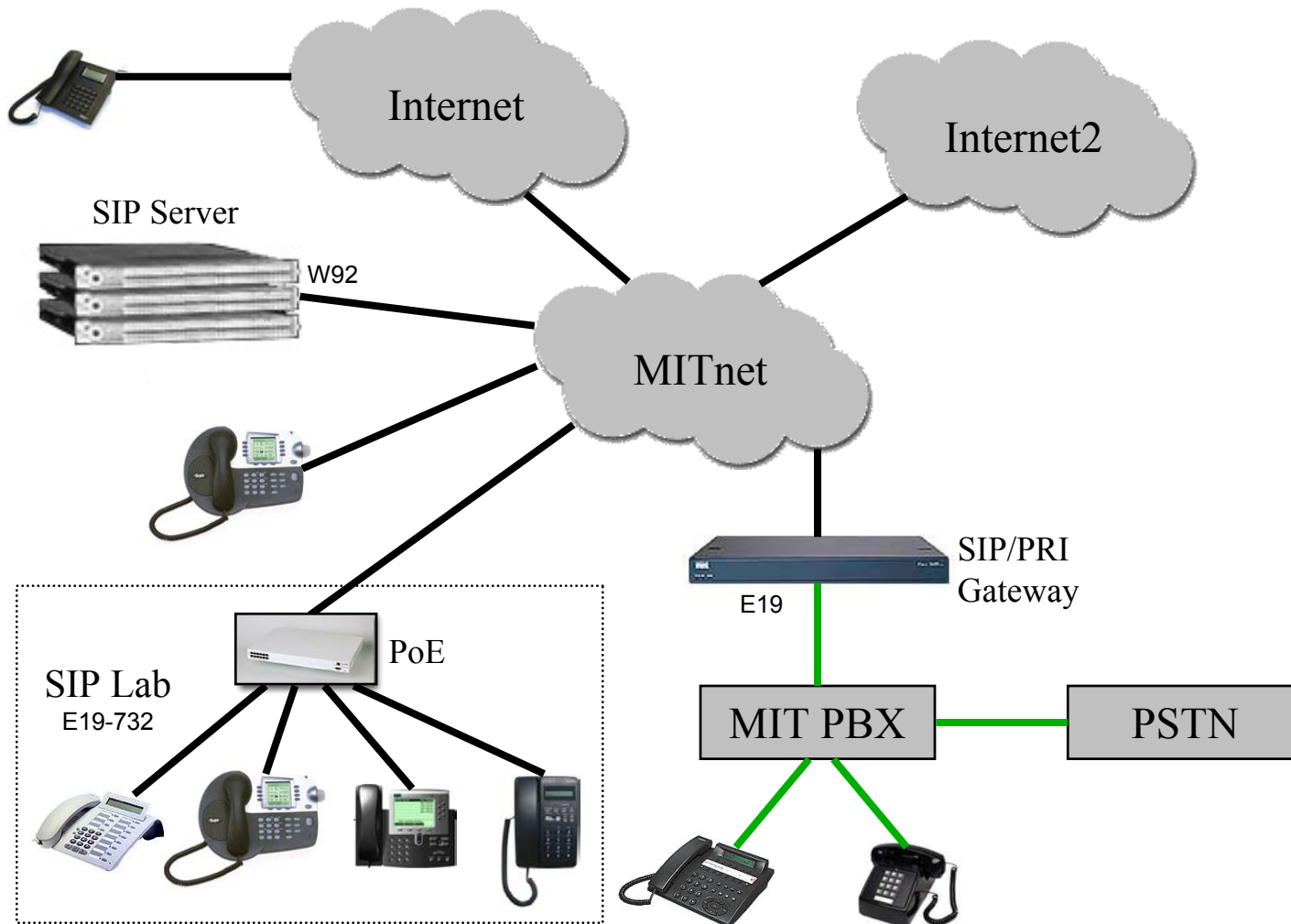
IS SIP Technology Trial

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- What it is:
 - SIP protocol evaluation
 - Integration issues
 - Interoperability issues
 - User evaluation/expectations
 - Identify applications/opportunities
 - System and equipment evaluation criteria
- What it isn't:
 - A production service
 - Vendor evaluation

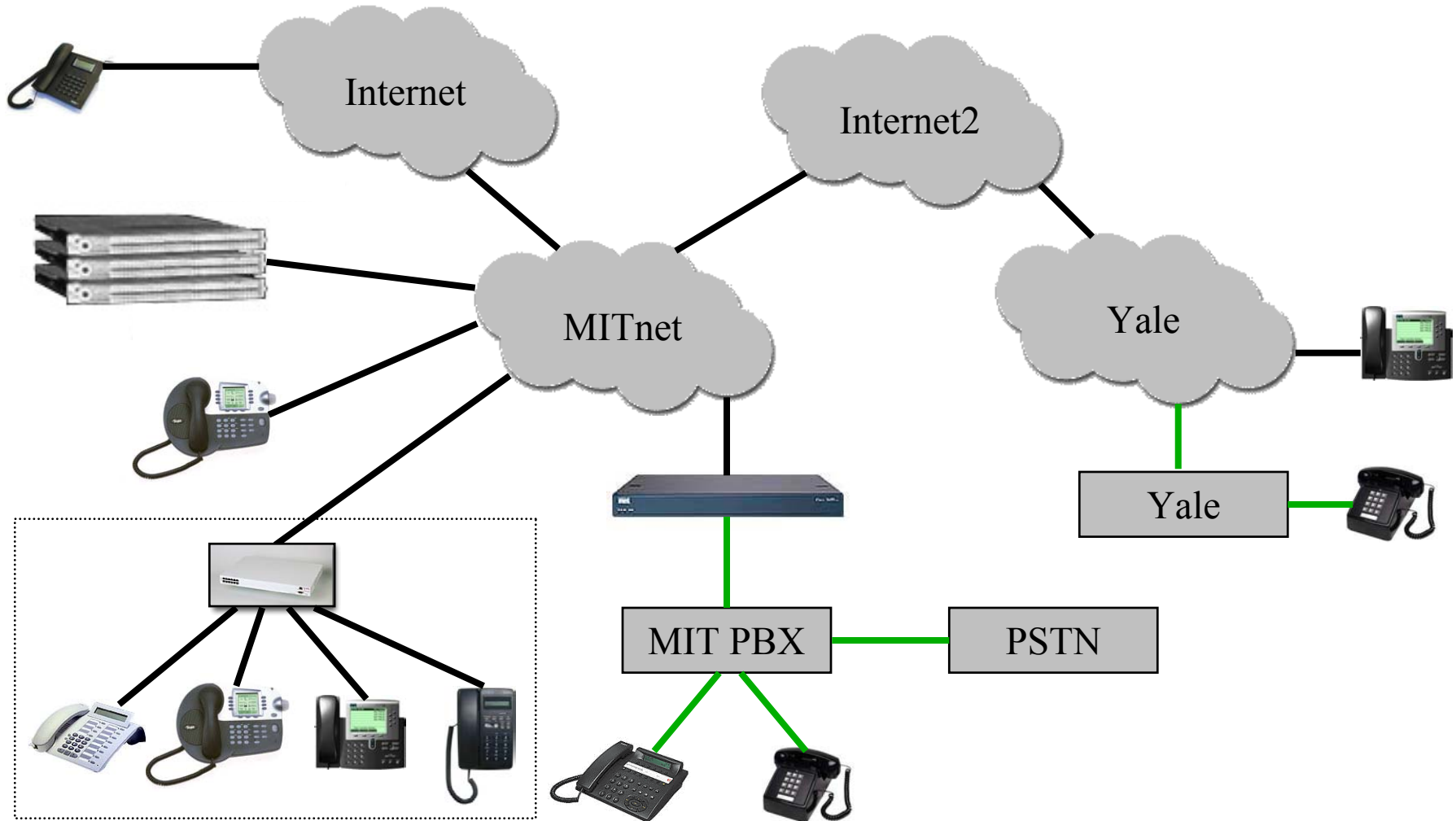
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Questions?



Telephony Future

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WARNING!



Telephony Future

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Telephony Future

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